



TEKNOLOGI
REKAYASA PERANGKAT
LUNAK

RPL524 - Kecerdasan Bisnis

INTRODUCTION OF BUSINESS INTELLIGENCE

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Education Background

S1 Teknik Informatika UIN Sultan Syarif Kasim Riau

S2 Teknik Informatika UPI YPTK Padang

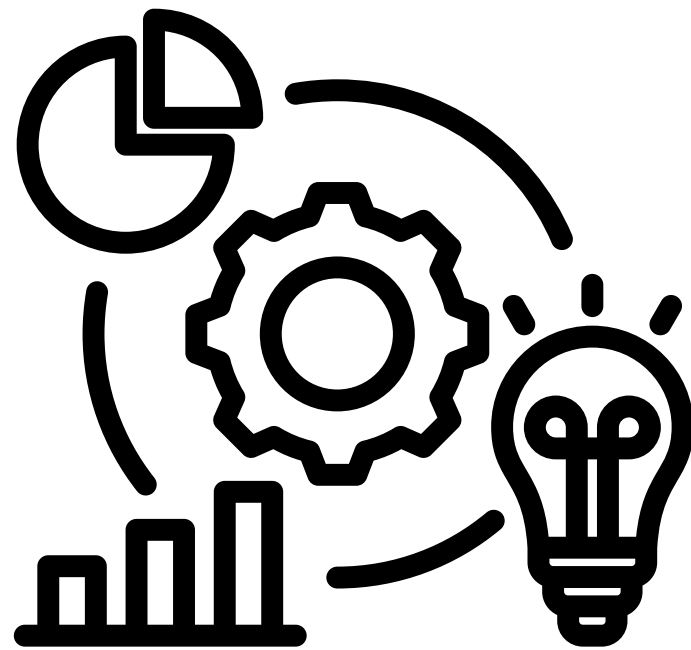
Homebase : D4 Teknologi Rekayasa Perangkat Lunak



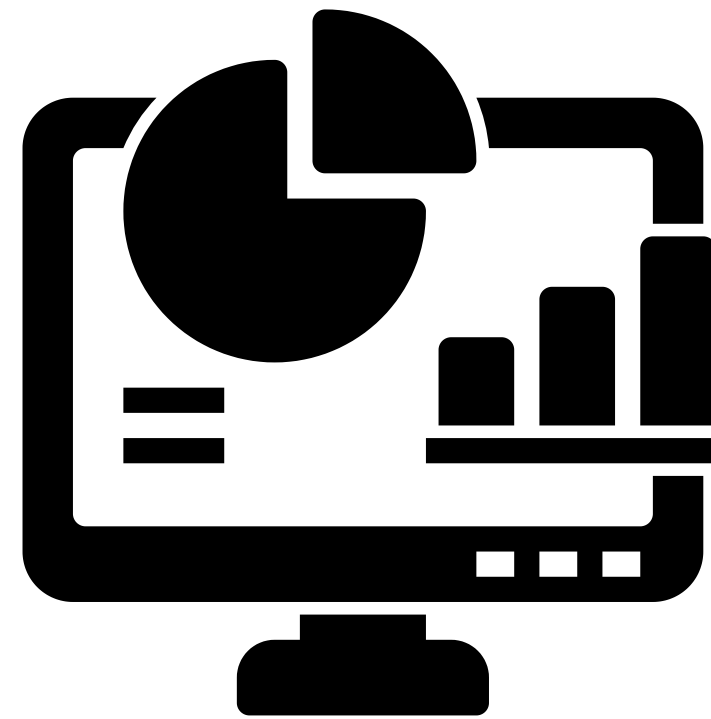
WHAT IS BUSINESS INTELLIGENCE?



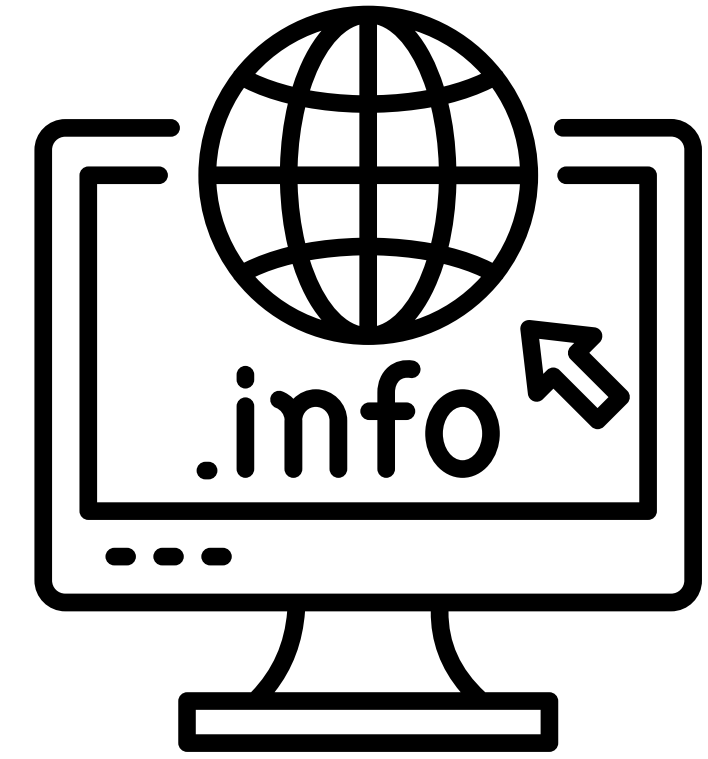
BUSINESS INTELLIGENCE



set of processes,
technology and tools



collect, process
analyze, and present data

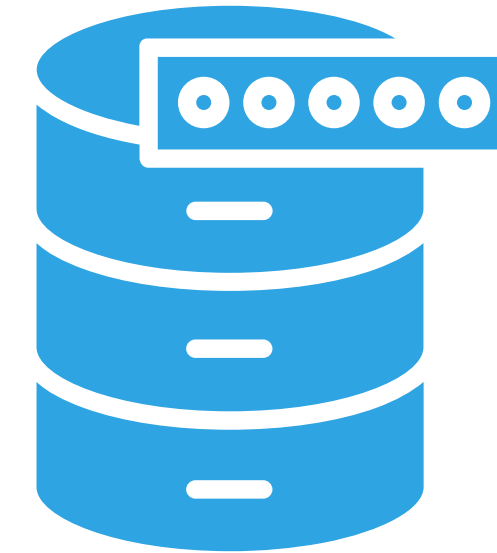


information for business
decision making

DATA



where do we
get it from?



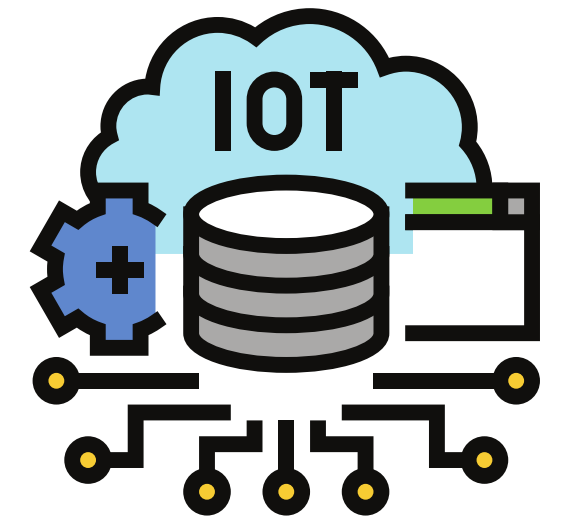
Company Database



Transaction

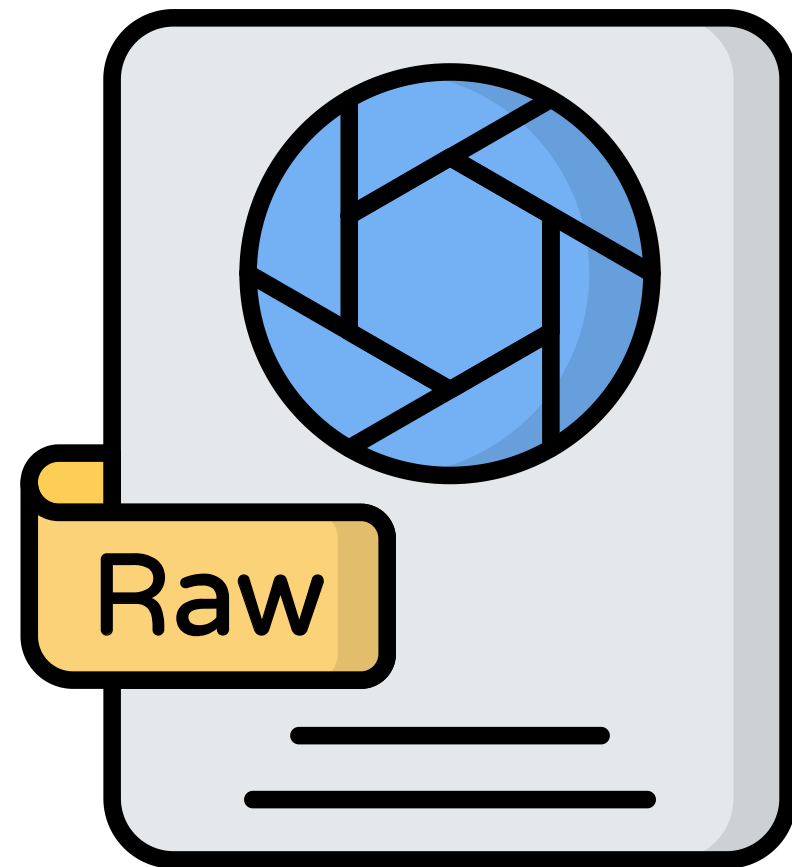


Social Media

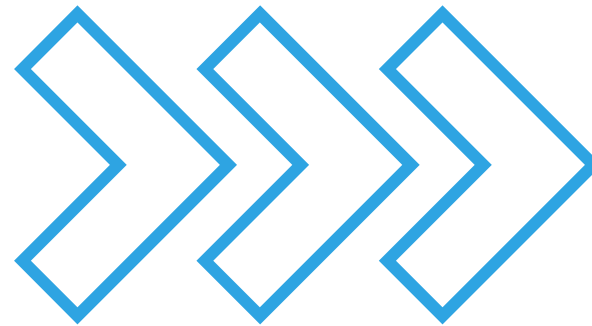


Internet of Things

BUSINESS INTELLIGENCE

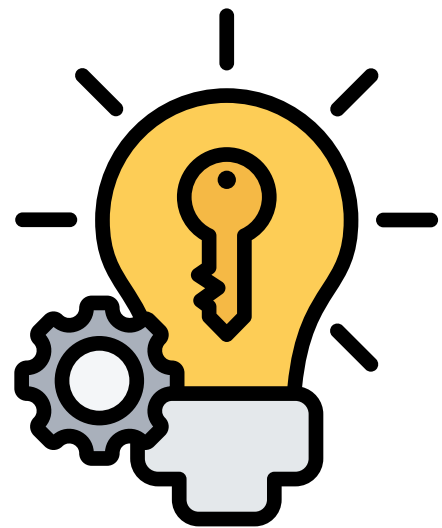


Raw Data



Insight

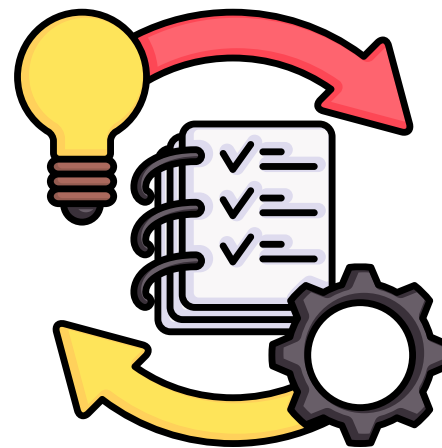
WHAT WE LEARN ABOUT?



Principles



Concept



Implementation

BUSINESS INTELLIGENCE



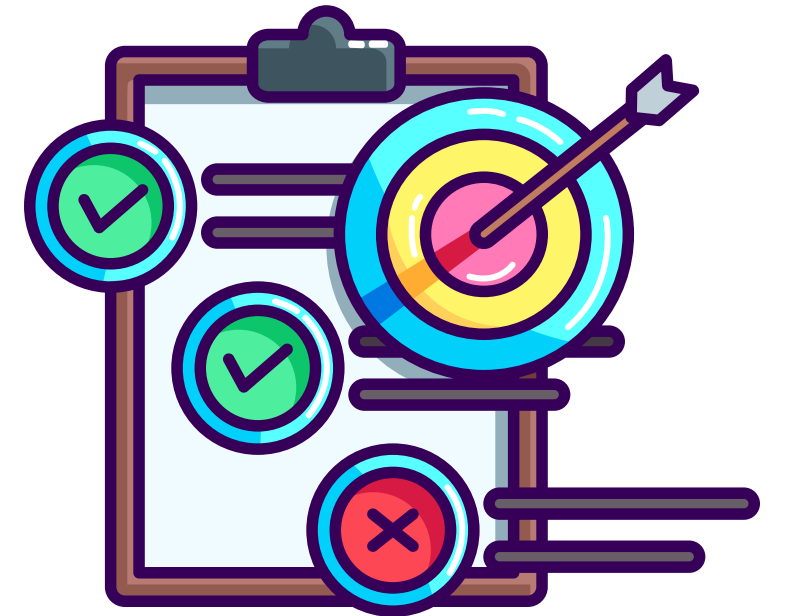
TECHNOLOGIES SUPPORTING BI

Business intelligence (BI) relies on several key technologies and supporting infrastructure. These are the main components:

- **Data Warehouse (DW) / Data Lake:** A data warehouse is a central repository of integrated data from one or more disparate sources. It stores current and historical data in one single place used for creating analytical reports. A data lake is a centralized repository that allows you to store all your data at any scale, whether it's structured, semi-structured, or unstructured.
- **ETL (Extract, Transform, Load) / ELT (Extract, Load, Transform) Tools:** These tools are used to extract data from various sources, transform it into a format suitable for analysis, and then load it into a data warehouse or data lake.
- **OLAP (Online Analytical Processing):** OLAP enables rapid, multi-dimensional analysis of large amounts of data stored in a data warehouse. This technology allows users to view data from different perspectives, such as by product, region, or time period.
- **Data Mining:** This technology involves discovering patterns, correlations, and anomalies in large datasets to predict future outcomes.
- **Reporting and Visualization Tools:** These are software applications that take processed data and present it in an easily understandable format, such as dashboards, charts, and graphs. Examples include Tableau, Power BI, and QlikView.

LEARNING OUTCOMES

- Basic concepts of business intelligence
- history and objectives of business intelligence
- organizational decision-making processes and levels
- Decision Support System using Data Mining Techniques
- Data visualization techniques



COMPARISON

ASPECT	WITH BI	WITHOUT BI
Desicion Making	Fast, accurate, data-driven	Intuition based, slow, error prone
Operational Efficiency	Efficient, automated reporting	Manual, duplicate prone
Market & Customer Analysis	Undestand trends & customer behaviour	Limited info, slow response
Competitive Advantage	Adaptive, quick to seize oppurtunity	Reactive, lagging behind competitor
Transparency & Accountability	Real-time monitoring, objective	Hard to evaluate performance

KPI ANALYSIS & PERFORMANCE METRICS

- **KPI (Key Performance Indicators)** → strategic indicators that show how well an organization is achieving its main objectives. KPIs are directly tied to business strategy.
- **Performance metrics** → quantitative measures of activities/operations. Not all metrics are KPIs, but every KPI is a metric.

Examples

- **Metric:** number of website visitors, units sold, average response time in customer service.
- **KPI:** sales conversion rate, net profit margin, customer retention rate.



QUESTION?

THANK YOU

Sukma Evadini, S.T., M.Kom
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